Vinayak Manoj Nair

Aerospace Engineering, Guidance, Navigation & Control · Flight Software · Systems Engineering

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01 Summary

Aerospace Engineering student at Texas A&M University (B.S. May 2027, Fast Track BS/MS) working across guidance, navigation and control, embedded flight software, structural analysis and systems engineering. Builds systems with verification evidence attached: hardware-in-the-loop test campaigns, Monte Carlo dispersion analysis, filter consistency validation and mesh-converged FEA. Currently supporting four ISS-bound experiments through NASA design review gates.

02 Education

Texas A&M University, B.S. Aerospace Engineering

College Station, TX · Expected May 2027 · GPA 3.45 / 4.00

Texas A&M University, M.S. Aerospace Engineering (Fast Track BS/MS)

In progress · Focus: guidance, navigation & control and spacecraft systems

03 Experience

Spacecraft Systems Engineer, TAMU-SPIRIT Capstone

Aegis Aerospace / ISS Payload Program · College Station, TX · Jan 2026 to Present

- Maintained 100% milestone compliance for 4 ISS-bound experiments targeting a Sep 2028 SpaceX Dragon launch by managing engineering change requests and NASA compliance tracking through SCR, SRR and PDR government design reviews.
- Reduced thermal design risk for the DASH payload by performing material selection trades and thermal analysis across −120 °C to +120 °C LEO cycling.
- Advanced the project toward late-2026 hardware manufacturing by coordinating quality documentation across an 8-person engineering team.

Controls & Embedded Systems Engineer

BLNC Turtle Robotics · College Station, TX · Aug 2025 to Dec 2025

- Reduced motor validation errors by standardizing bench-test procedures in LabVIEW and maintaining as-built documentation.
- Achieved full autonomous stabilization of a two-wheeled inverted-pendulum rover, building hardware from component level and implementing cascaded PID control.

Undergraduate Research Assistant

Texas A&M ACE Lab, Dr. Sasangohar · College Station, TX · Dec 2023 to Dec 2024

- Cut manual analysis time by 40% and accelerated report delivery by 2 days by building a Python pipeline processing 5,000+ data points per run.

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04 Selected projects

GHOST: GPS-Denied Occlusion-Survivable Tracker ACTIVE

C++ · Python · ESKF · IMM · PX4 SITL · MAVLink · Raspberry Pi 4B

- Built a dual-filter visual-inertial estimation pipeline: a 9-state error-state Kalman filter fusing 200 Hz IMU propagation with 30 Hz AprilTag pose fixes, plus a CV/CTRV interacting-multiple-model target tracker with χ^2 innovation gating.
- Iterated the design through 10 documented revisions with 20 engineering flaws identified and fixed via external review; implemented NIS consistency validation and continuous integration.

AstraSim-FSW: Flight Software HIL Verification Framework HIL VERIFIED

C++17 · Python · CMake / CTest · UDP · CRC-16-CCITT · Raspberry Pi aarch64

- Built a flight-software-in-the-loop verification framework with binary UDP command/telemetry, CRC-16 packet validation, command acknowledgement, fault injection, watchdog and health-monitor logic, YAML scenarios and Monte Carlo regression.
- Achieved 9/9 CTest suites, 5/5 scenario runs and 25/25 Monte Carlo regressions passing on both x86-64 host and aarch64 Raspberry Pi target, with committed hardware evidence.
- Reduced packet validation overhead 40% by migrating from bitwise to table-driven CRC-16-CCITT, measured on the target rather than the development host.

Rocket Landing GNC Simulation

Python · NumPy / SciPy · 6-DOF · Monte Carlo · TVC

- Developed a 6-DOF booster descent simulator with aerothermodynamic heating, propulsion and mass-depletion modelling, and closed a PID thrust-vector control loop around estimated (not truth) state.
- Validated GNC robustness across 500 Monte Carlo stochastic wind runs to sub-2.0 m landing accuracy, and attributed the failure tail to low-altitude wind shear against actuator rate limits.

AeroFrame-MD11: CAD Assembly & Structural Analysis

SolidWorks · Abaqus FEA (static / thermal / modal) · GD&T

- Identified 3 structural failure sites under 2.5 g flight loading by modelling a 24-part aircraft assembly with full parametric CAD and GD&T, with mesh convergence used to separate real concentrations from singularities.
- Produced a complete manufacturing-release drawing package with fabrication callouts and bill of materials.

AI-Powered Autonomous Interception Robot

Python · YOLO · Kalman filter · PID · Raspberry Pi 4

- Achieved sub-100 ms closed-loop latency on embedded Linux by integrating YOLO object detection, Kalman filter state estimation and PWM motor control with per-stage latency instrumentation.
- Compensated residual pipeline latency by propagating the state estimate to actuation time, converting a systematic tracking lag into bounded error.

05 Technical skills

PROGRAMMING	C++17, Python, MATLAB, LabVIEW, Bash, CMake
GNC & ESTIMATION	Error-state Kalman filter (ESKF), extended and linear Kalman filtering, IMM (CV/CTRV) tracking, proportional navigation, PID and cascaded PID, thrust-vector control, sensor fusion
SIMULATION	6-DOF flight dynamics, Monte Carlo dispersion analysis, PX4 SITL, Gazebo, MAVLink, aerothermodynamics
CAD & FEA	SolidWorks parametric assemblies, GD&T, drawing release packages, Abaqus static/thermal/modal analysis, mesh convergence, margin of safety
EMBEDDED SYSTEMS	Embedded Linux (aarch64), Raspberry Pi, binary command/telemetry protocols, CRC-16-CCITT, watchdog and FDIR logic, IMU and camera integration, PWM control
TESTING & V&V	Hardware-in-the-loop verification, fault injection, CTest regression, NIS filter-consistency validation, bench-test procedure standardisation, evidence capture
SYSTEMS ENGINEERING	Requirements decomposition and traceability, SCR/SRR/PDR design reviews, engineering change requests, interface control definitions, trade studies, V&V planning
TOOLS & PLATFORMS	Linux, Git/GitHub, continuous integration, SSH remote targets, NumPy/SciPy, OpenCV

06 Availability

SEEKING	Summer 2026 internship; May 2027 full-time
TARGET ROLES	GNC engineering, flight software, embedded systems, simulation and analysis, systems engineering, test & verification
LOCATION	College Station, TX, open to relocation
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